

**BUSINESS REPORT – PREDICTIVE MODELLING
PROJECT**

**Casestudy -
OTT Media service**

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PROBLEM STATEMENT:

Context:

An over-the-top (OTT) media service is a media service offered directly to viewers via the internet. The term is most synonymous with subscription-based video-on-demand services that offer access to film and television content, including existing series acquired from other producers, as well as original content produced specifically for the service. They are typically accessed via websites on personal computers, apps on smartphones and tablets, or televisions with integrated Smart TV platforms.

Presently, OTT services are at a relatively nascent stage and are widely accepted as a trending technology across the globe. With the increasing change in customers' social behavior, which is shifting from traditional subscriptions to broadcasting services and OTT on-demand video and music subscriptions every year, OTT streaming is expected to grow at a very fast pace. The global OTT market size was valued at USD 121.61 billion in 2019 and is projected to reach \$1,039.03 billion by 2027, growing at a CAGR of 29.4% from 2020 to 2027. The shift from television to OTT services for entertainment is driven by benefits such as on-demand services, ease of access, and access to better networks and digital connectivity.

With the outbreak of COVID-19, OTT services are striving to meet the growing entertainment appetite of viewers, with some platforms already experiencing a 46% increase in consumption and subscriber count as viewers seek fresh content. With innovations and advanced transformations, which will enable the customers to access everything they want in a single space, OTT platforms across the world are expected to increasingly attract subscribers on a concurrent basis.

Objective:

ShowTime is an OTT service provider and offers a wide variety of content (movies, web shows, etc.) for its users. They want to determine the driver variables for first-day content viewership so that they can take necessary measures to improve the viewership of the content on their platform. Some of the reasons for the decline in viewership of content would be the decline in the number of people coming to the platform, decreased marketing spend, content timing clashes, weekends, and holidays, etc. They have hired you as a Data Scientist, shared the data of the current content on their platform, and asked you to analyze the data and come up with a linear regression model to determine the driving factors for first-day viewership.

Data Description:

The data contains the different factors to analyze for the content. The detailed data dictionary is given below.

Data Dictionary:

- visitors: Average number of visitors, in millions, to the platform in the past week
- ad_impressions: Number of ad impressions, in millions, across all ad - campaigns for the content (running and completed)
- major_sports_event: Any major sports event on the day
- genre: Genre of the content
- dayofweek: Day of the release of the content
- season: Season of the release of the content
- views_trailer: Number of views, in millions, of the content trailer
- views_content: Number of first-day views, in millions, of the content

The following questions need to be answered as part of the EDA section of the project:

1. What does the distribution of content views look like?
2. What does the distribution of genres look like?
3. The day of the week on which content is released generally plays a key role in the viewership. How does the viewership vary with the day of release?
5. How does the viewership vary with the season of release?
6. What is the correlation between trailer views and content views?

Key Point to Analyse:

Analyze the data and come up with a linear regression model to determine the driving factors for first-day viewership.

MODULES DEVELOPED:

Python libraries used	Environment
Numpy, Pandas, Matplotlib, Seaborn, Scipy.stats, sklearn.model_selection, sklearn.metrics, statsmodels.api, statsmodels.stats.outliers_influence, sklearn.linear_model	Google colab

DATA OVERVIEW:

- The dataset contains information about OTT Content.
- Data is checked with trailer views and content views and an analysis based on the season and genre is done.
- The data contains 8 attributes and 1000 observations.
- There are 3 categorical variables and 5 numerical variables in the data
- The target variable is views_content of ottdata which is float type.

	visitors	ad_impressions	major_sports_event	genre	dayofweek	season	views_trailer	views_content
0	1.67	1113.81	0	Horror	Wednesday	Spring	56.70	0.51
1	1.46	1498.41	1	Thriller	Friday	Fall	52.69	0.32
2	1.47	1079.19	1	Thriller	Wednesday	Fall	48.74	0.39
3	1.85	1342.77	1	Sci-Fi	Friday	Fall	49.81	0.44
4	1.46	1498.41	0	Sci-Fi	Sunday	Winter	55.83	0.46

Fig1. First five rows of data

STATISTICAL SUMMARY:

- Statistical summary of five key variables i.e., visitors, ad impressions, major sports event, views trailer, and views content is shown.
- The average number is 1.70 with a standard deviation of 0.23.
- Higher visitor counts seem to coincide with higher ad impressions, suggesting that increased foot traffic leads to more ad views.
- Trailer views vary significantly from 30.08 to 199.92, suggesting that certain trailers pull more viewership.

	count	mean	std	min	25%	50%	75%	max
visitors	1000.0	1.70429	0.231973	1.25	1.5500	1.70	1.830	2.34
ad_impressions	1000.0	1434.71229	289.534834	1010.87	1210.3300	1383.58	1623.670	2424.20
major_sports_event	1000.0	0.40000	0.490143	0.00	0.0000	0.00	1.000	1.00
views_trailer	1000.0	66.91559	35.001080	30.08	50.9475	53.96	57.755	199.92
views_content	1000.0	0.47340	0.105914	0.22	0.4000	0.45	0.520	0.89

Fig2. Statistical analysis of data

EXPLORATORY DATA ANALYSIS

UNIVARIATE ANALYSIS OF NUMERICAL VARIABLES:

1.A histogram and boxplot analysis of views-trailer variable:

- Data is heavily right skewed with extreme outliers in it.

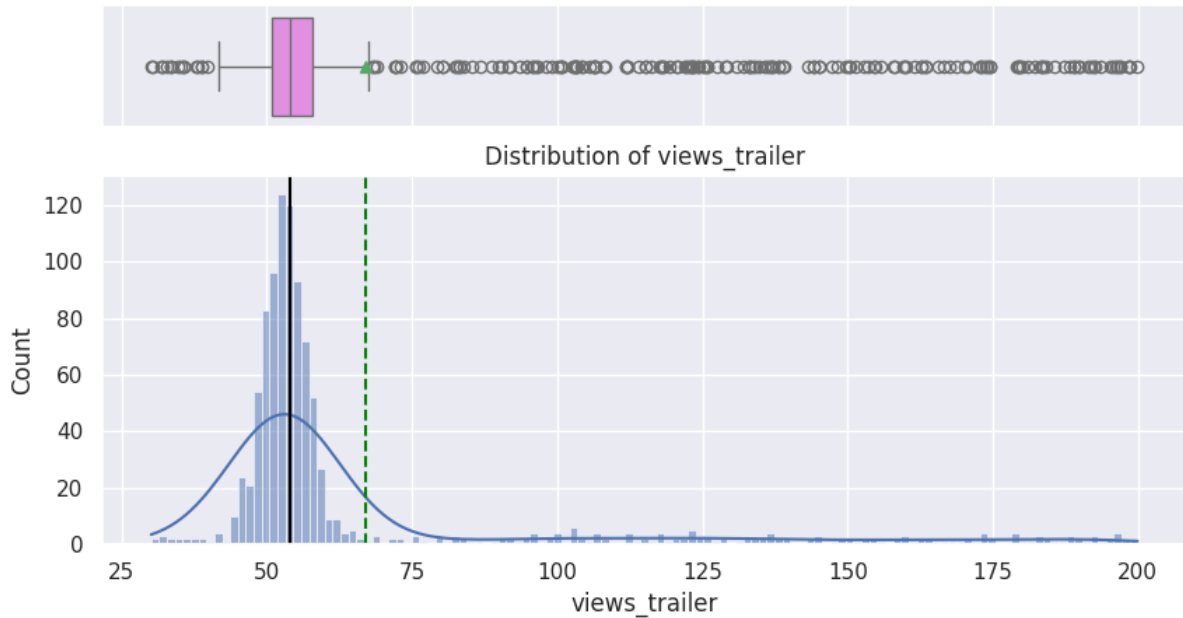


Fig3. Distribution of views trailer

2.A histogram and boxplot analysis of ad-impressions variable:

- Data is heavily right skewed with one outlier in it



Fig4. Distribution of ad_impressions.

3.A histogram and boxplot analysis of visitors variable:

- Data seems a almost normally distributed data with only one outlier datapoint.

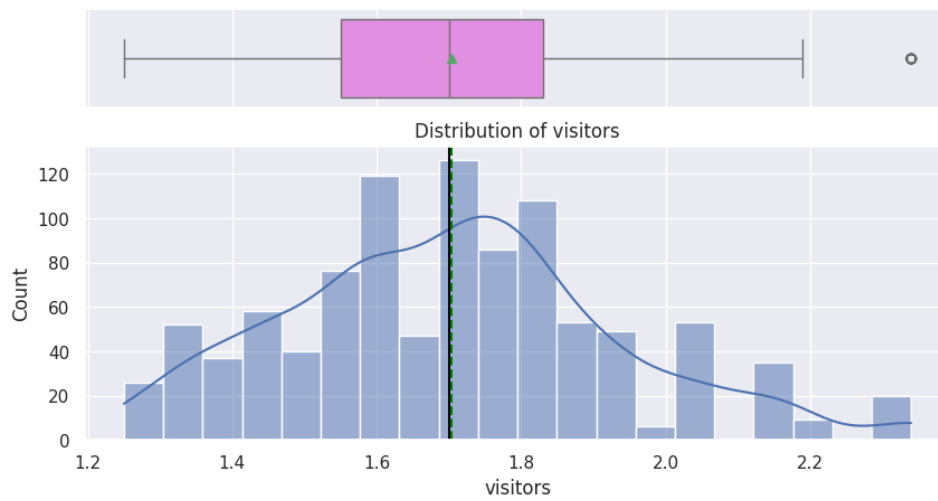


Fig5. Distribution of visitors

UNIVARIATE ANALYSIS OF CATEGORICAL VARIABLES:

Views content by dayofweek:

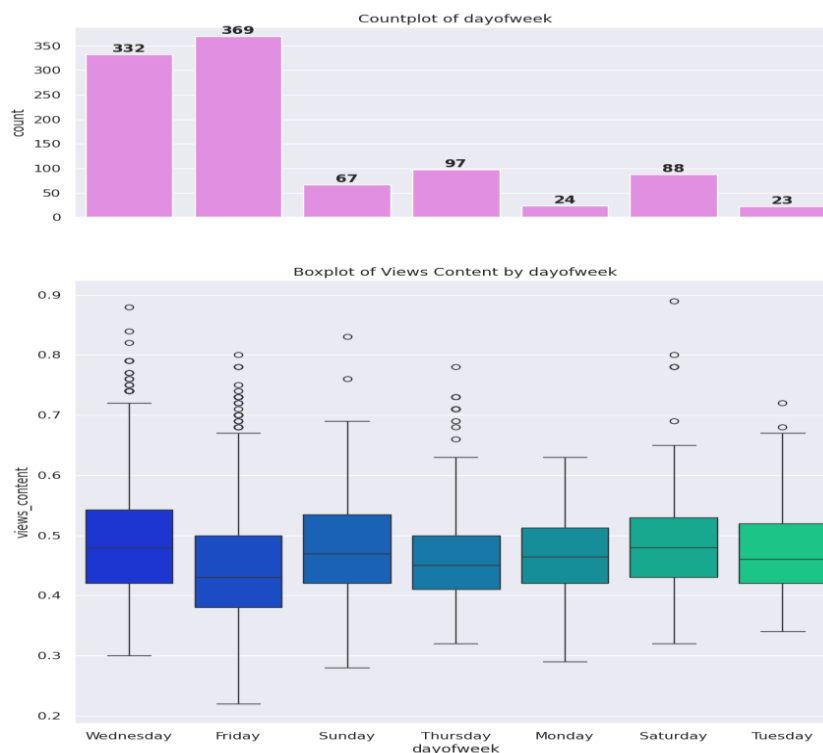


Fig6. Views content by dayofweek

- It shows a Friday and Wednesday has highest views.

Views content by season:

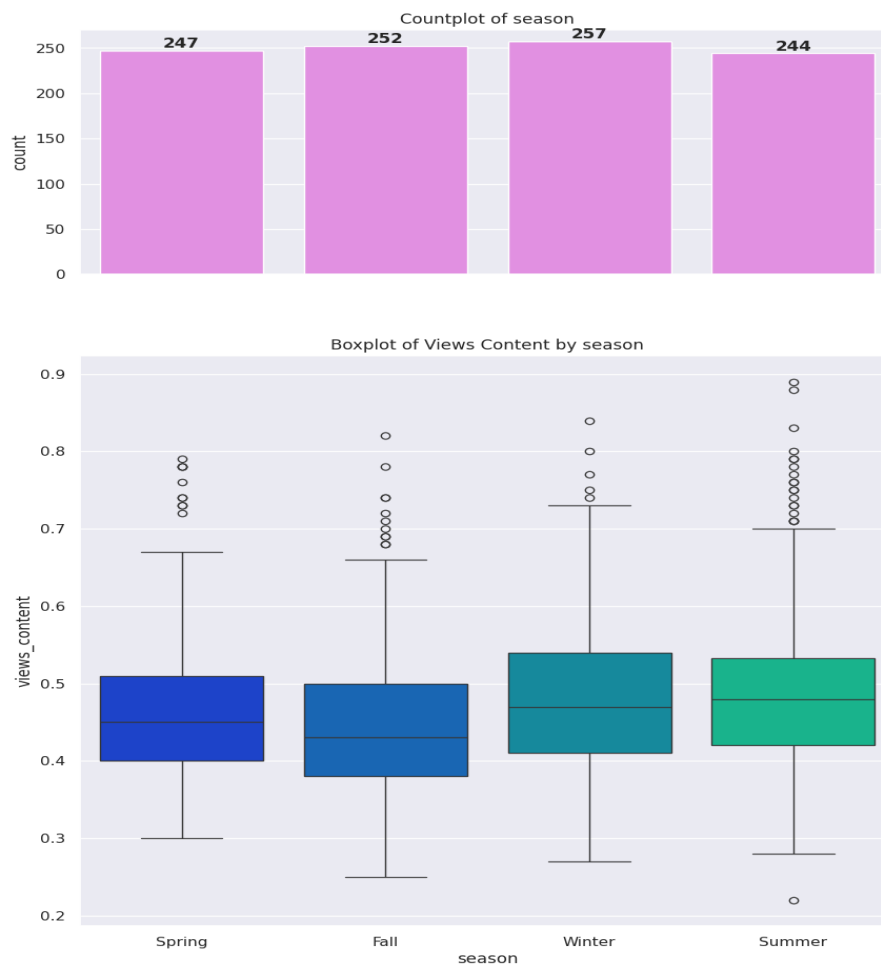


Fig7. Views content by season

- Summer season shows highest views.
- Data also shows extreme outliers with median around 0.4 and 0.5.

BIVARIATE ANALYSIS:

Analysis of visitors by views content:

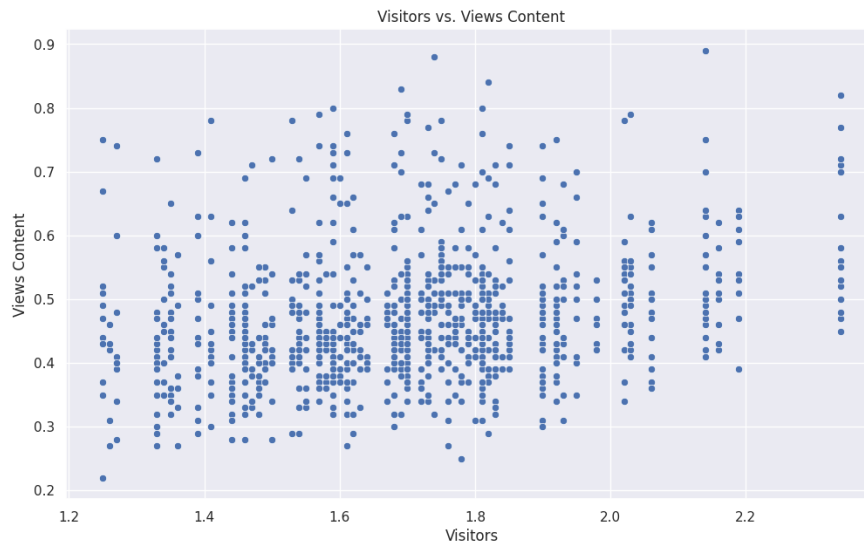


Fig 8. Visitors vs. views content

- Data shows as visitors increase views of the content also increase.

Comparing the views across different Genres:

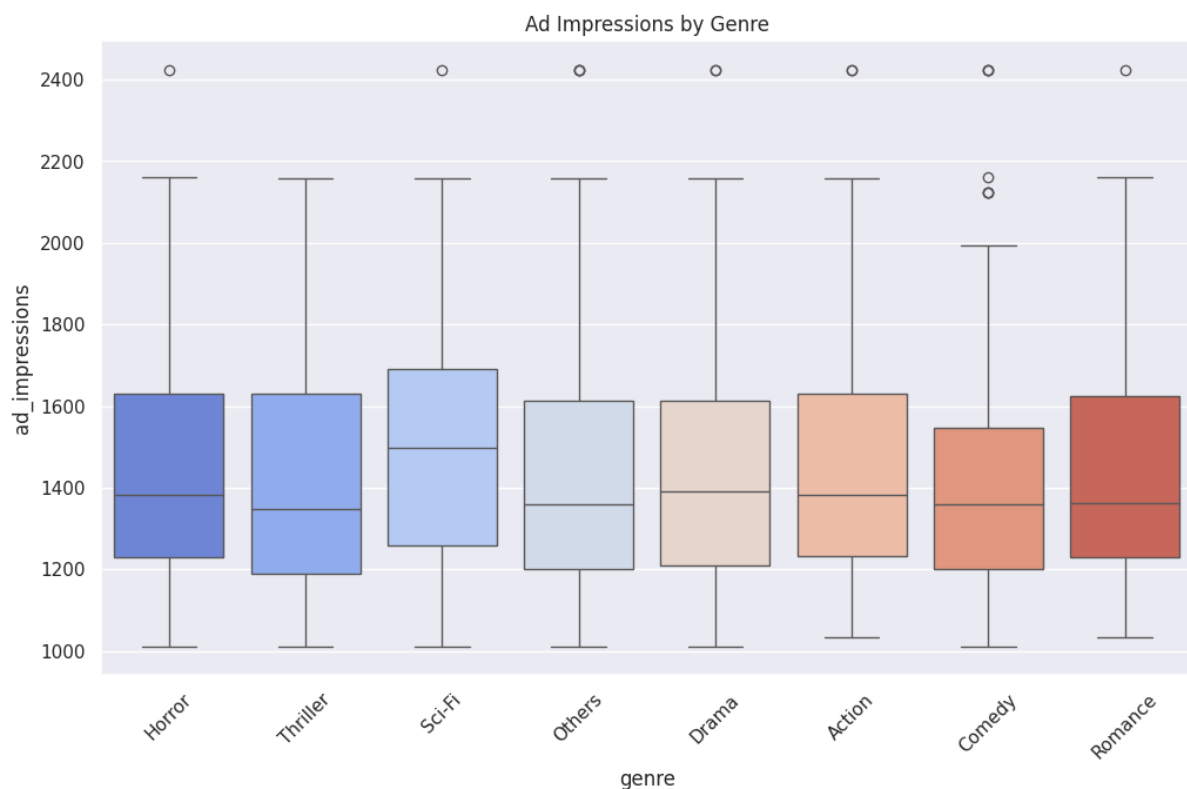


Fig 9. Ad Impressions by Genre

- Data shows scifi and romance genre has more popularity.

Comparing the visitors across different Genres.

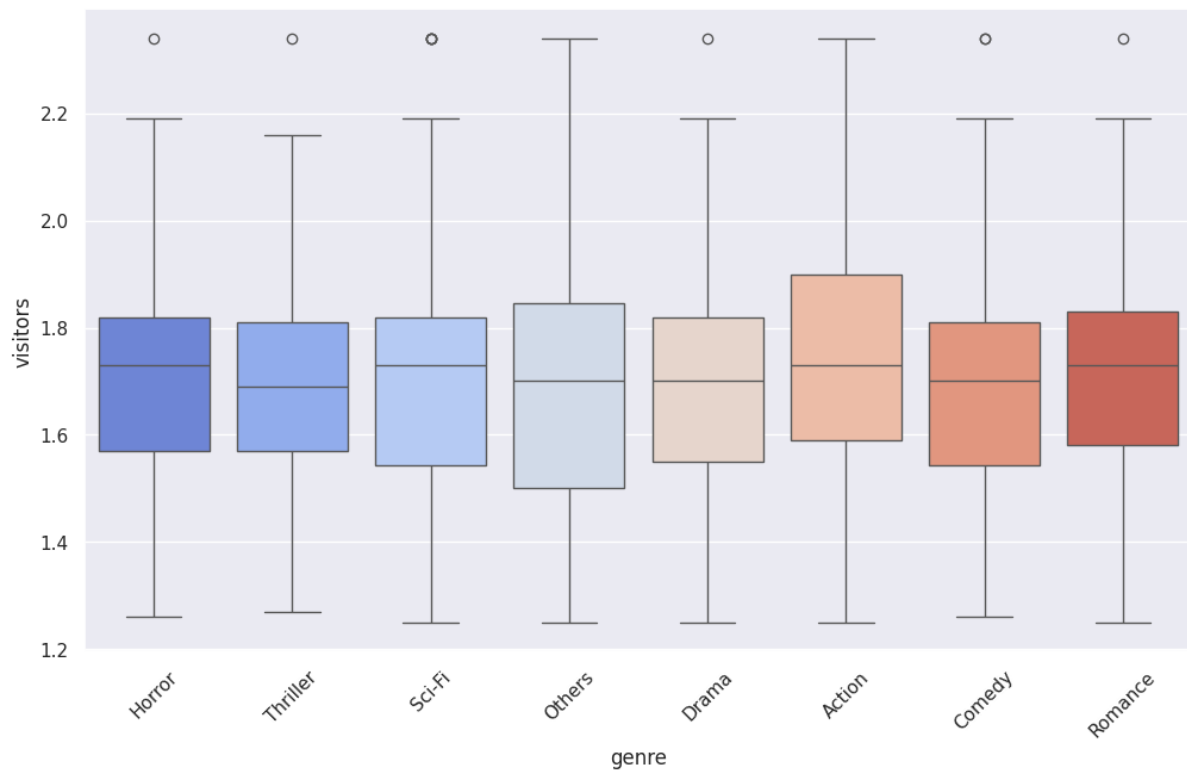


Fig 10. Comparing the visitors across different Genres

KEY QUESTIONS AND ANSWERS

Key questions 1: What does the distribution of content views look like?

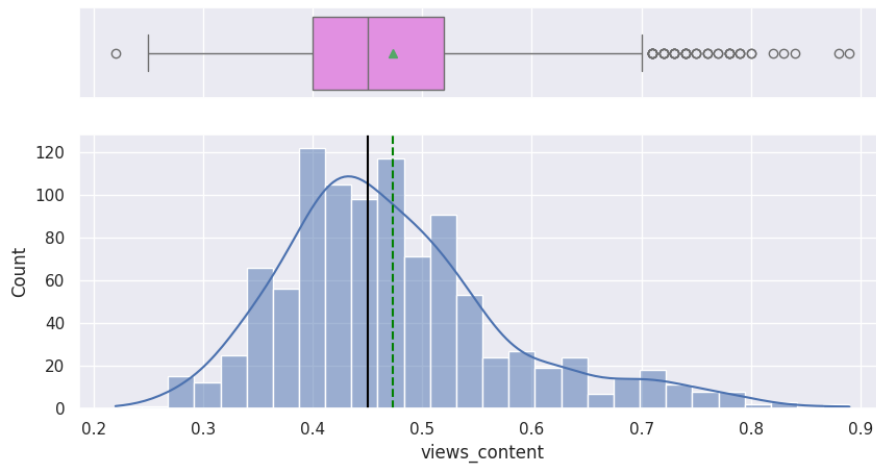


Fig11. Views content

Ans:

- The distribution of views content looks like a right_skewed data which indicates that first day views of some content are extremely high which is pulling the average while most of the other data points are low.
- This indicates that some content or show is more popular having better views than other
- Data looks like almost a bell curve but right skewed to some extent.

Key questions 2: What does the distribution of genres look like?

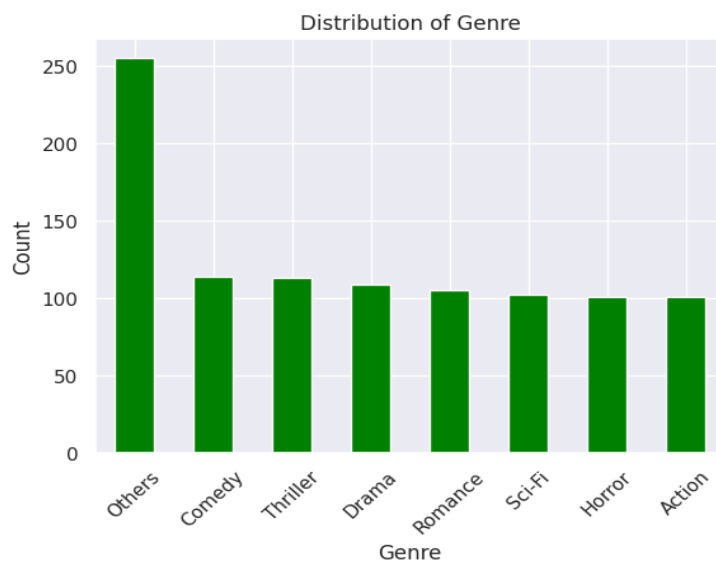


Fig12. Distribution of Genre

Ans: It shows a right skewed data which infers content in genre-'others' has more views

Key question 3: The day of the week on which content is released generally plays a key role in the viewership. How does the viewership vary with the day of release?

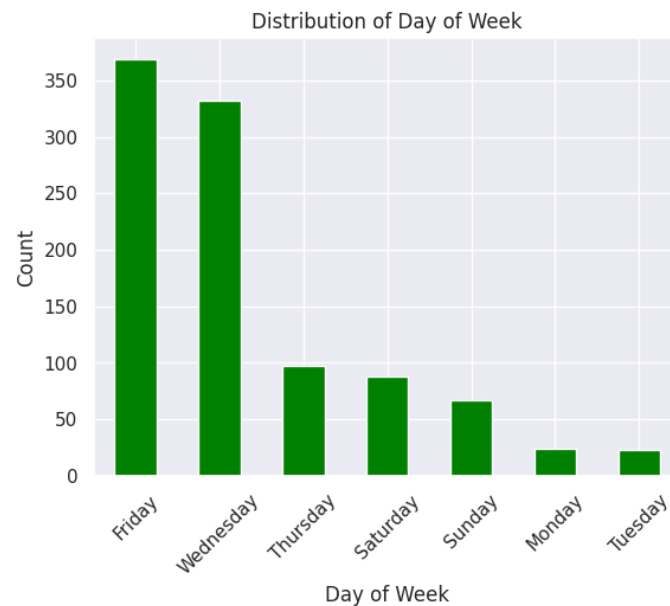


Fig.13. Distribution of dayofweek

Ans:

- First graph indicates that the weekends are the peakdays attracting higher views compared to weekdays.
- When looked at each day in the second graph, it states a wednesday and friday attracts better views compared to other days.

Key question 4: How does the viewership vary with the season of release?

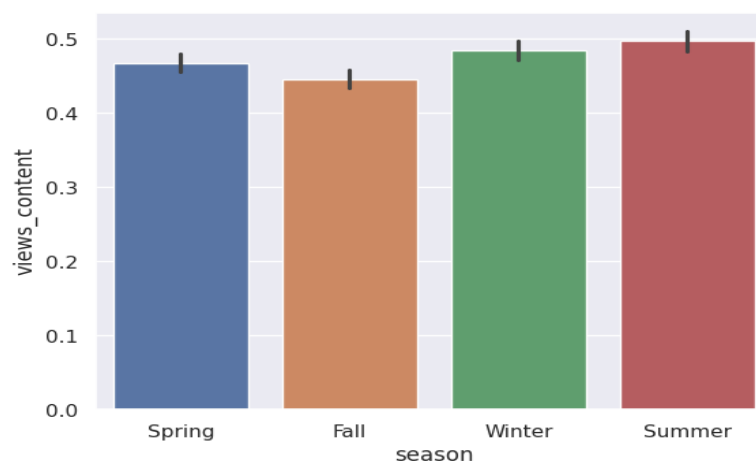


Fig 14. Viewership by season

Ans:

- Summer attracts more viewers compared to any other seasons. There is no significant fluctuations in the viewership when compared amongst the seasons.

Key question 5: What is the correlation between trailer views and content views?

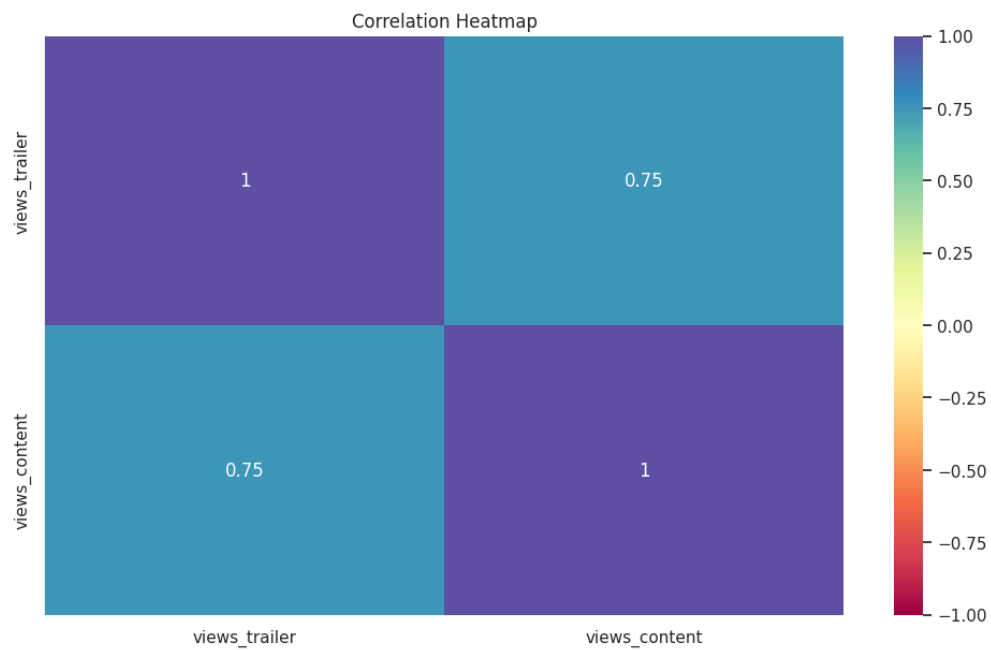


Fig 15. Correlation heatmap of views content and views trailer

Ans:

- The above heatmap reveals a strong correlation between trailer views and content views, as it shows a 0.75 correlation which is nearer to 1.

DATA PREPROCESSING

- **Treating duplicates:** There are no duplicates in the data.
- **Treating missing values:** There are no missing values in the data.
- **Feature engineering and one-hot encoding:** We did a feature engineering of the data by converting genre, dayofweek, and season into numerical representations using one-hot encoding or label encoding.
- **Feature engineering:** Also, we converted major_sports_event into a binary feature (already exists as 0 and 1)
- **Outlier treatment:** There are quite a few outliers in the data however, we will not treat them as they are proper values

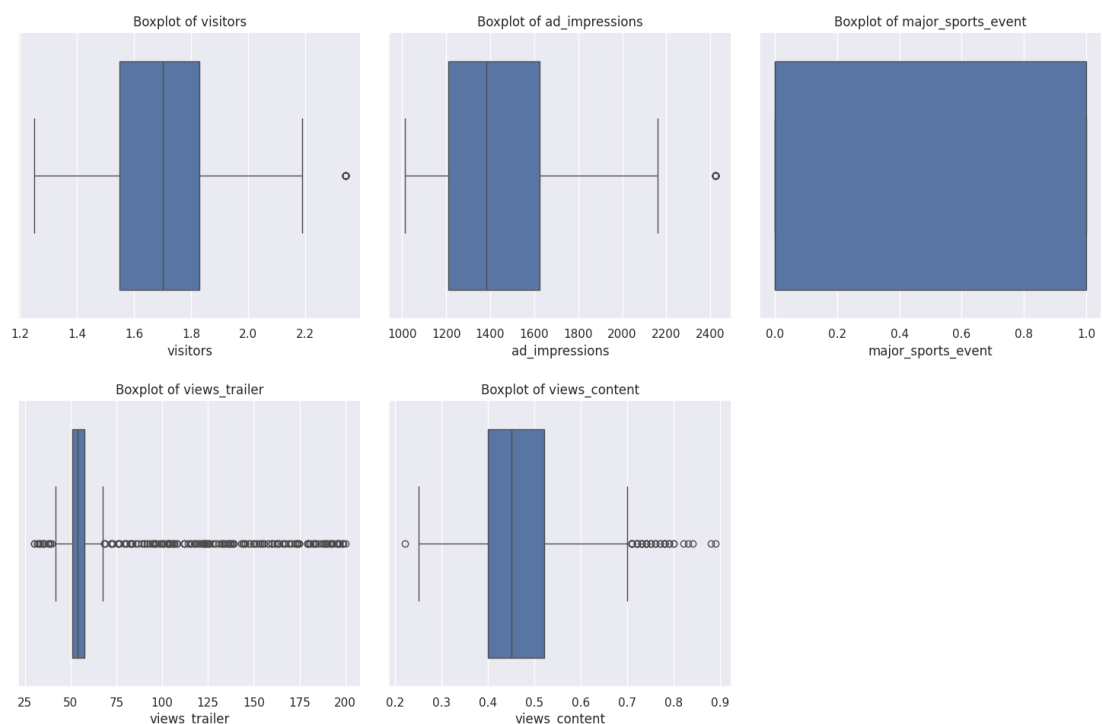


Fig 16. Boxplot of numerical variables

DATA PREPARATION FOR MODELLING:

- We defined X & Y variables and assumed the target variable as views content.
- Since data will not have a constant, we added it for modelling purpose to add an intercept.
- We created dummy variables of the X and added them to the data and changed the datatype as float.
- And we split the data into train and test data into 70:30 ratio for proceeding with building a linear regression model.

Number of rows in train data = 700

Number of rows in test data = 300

INTERPRETING THE LINEAR REGRESSION RESULTS FOR 'OLSMODEL'

1. **Adjusted. R-squared:** It reflects the fit of the model.
 - Adjusted R-squared values generally range from 0 to 1, where a higher value generally indicates a better fit, assuming certain conditions are met.
 - In our case, the value for adj. R-squared is **0.788**, which is good.
2. **const coefficient:** It is the Y-intercept.
 - It means that if all the predictor variable coefficients are zero, then the expected output (i.e., Y) would be equal to the *const* coefficient.
 - In our case, the value for const coefficient is **-0.2611**
3. **Coefficient of a predictor variable:** It represents the change in the output Y due to a change in the predictor variable (everything else held constant).
 - In our case, the coefficient of visitors is **0.1287**.

MODEL PERFORMANCE CHECK FOR OLSMODEL:

We check the performance of the model using different metrics.

- We will be using metric functions defined in sklearn for RMSE, MAE, and R2 .
- We will define a function to calculate MAPE and adjusted R2 .
- The mean absolute percentage error (MAPE) measures the accuracy of predictions as a percentage, and can be calculated as the average absolute percent error for each predicted value minus actual values divided by actual values. It works best if there are no extreme values in the data and none of the actual values are 0.

Observations of OLSMODEL:

- The training R2 is 0.79, so the model is not underfitting
- The train and test RMSE and MAE are comparable, so the model is not overfitting either
- MAE suggests that the model can predict ottdata within a mean error of 0.04 on the test data
- MAPE of 9.03 on the test data means that we can predict within 9% of the ottdata.

CHECKING FOR LINEAR REGRESSION ASSUMPTIONS:

We will be checking the following Linear Regression assumptions:

1. No Multicollinearity
2. Linearity of variables
3. Independence of error terms
4. Normality of error terms
5. No Heteroscedasticity

1. TEST FOR MULTICOLLINEARITY

Multicollinearity occurs when predictor variables in a regression model are correlated which is a problem because predictor variables should be independent. If the correlation between variables is high, it can cause problems when we fit the model and interpret the results.

One way to test it is by using the Variance Inflation Factor, or VIF.

Variance Inflation Factor (VIF): Variance inflation factors measure the inflation in the variances of the regression parameter estimates due to collinearities that exist among the predictors.

General Rule of thumb:

If VIF is between 1 and 5, then there is low multicollinearity.

If VIF is between 5 and 10, we say there is moderate multicollinearity.

If VIF is exceeding 10, it shows signs of high multicollinearity.

Observations:

- There are two columns `views_trailer` and `views_trailer_transformed` which show high multicollinearity which is $VIF > 5$, hence we can drop these columns.

Multicollinearity: To remove multicollinearity

1. Drop every column one by one that has a VIF score greater than 5.
2. Look at the adjusted R-squared and RMSE of all these models.
3. Drop the variable that makes the least change in adjusted R-squared.
4. Check the VIF scores again.
5. Continue till you get all VIF scores under 5.

Observation:

- We have decided to drop `views_trailer_transformed` column and check the VIF again.
- After performing this step, we have dealt with multicollinearity and observe clean data.

Let us rebuild the model using the updated set of predictors variables

OBSERVATIONS OF 'OLSMODEL1':

- We can see that adj. R-squared has dropped from 0.788 to 0.785, which shows that the dropped columns did not have much effect on the model
- As there is no multicollinearity, we can look at the p-values of predictor variables to check their significance

DEALING WITH HIGH P-VALUE VARIABLES:

- Some of the dummy variables in the data have p-value > 0.05. So, they are not significant and we will drop them
- But sometimes p-values change after dropping a variable. So, we will not drop all variables at once
- Instead, we will do the following:
 - Build a model, check the p-values of the variables, and drop the column with the highest p-value
 - Create a new model without the dropped feature, check the p-values of the variables, and drop the column with the highest p-value
 - Repeat the above two steps till there are no columns with p-value > 0.05

OBSERVATIONS OF OLSMODEL2:

- P values > 0.05 are now addressed. so, we will consider the features in `x_train3` as the final set of predictor variables and `olsmod2` as the final model.
- The adjusted R-squared is 0.786, i.e., our model can explain ~78.6% of the variance
- The adjusted R-squared in `olsmod1` (where we considered the variables without multicollinearity) was 0.785
- This shows that the variables we dropped were not affecting the model
- RMSE and MAE values are comparable for train and test sets, indicating that the model is not overfitting.

Now we will check the rest of the assumptions on *olsmod2*.

- Linearity of variables
- Independence of error terms
- Normality of error terms
- No Heteroscedasticity

2. TEST FOR LINEARITY AND INDEPENDENCE

Why the test?

- Linearity describes a straight-line relationship between two variables; predictor variables must have a linear relation with the dependent variable.

To check linearity and independence:

- Make a plot of fitted values vs residuals.
- If they do not follow any pattern, then we say the model is linear and residuals are independent.
- Otherwise, the model is showing signs of non-linearity and residuals are not independent.

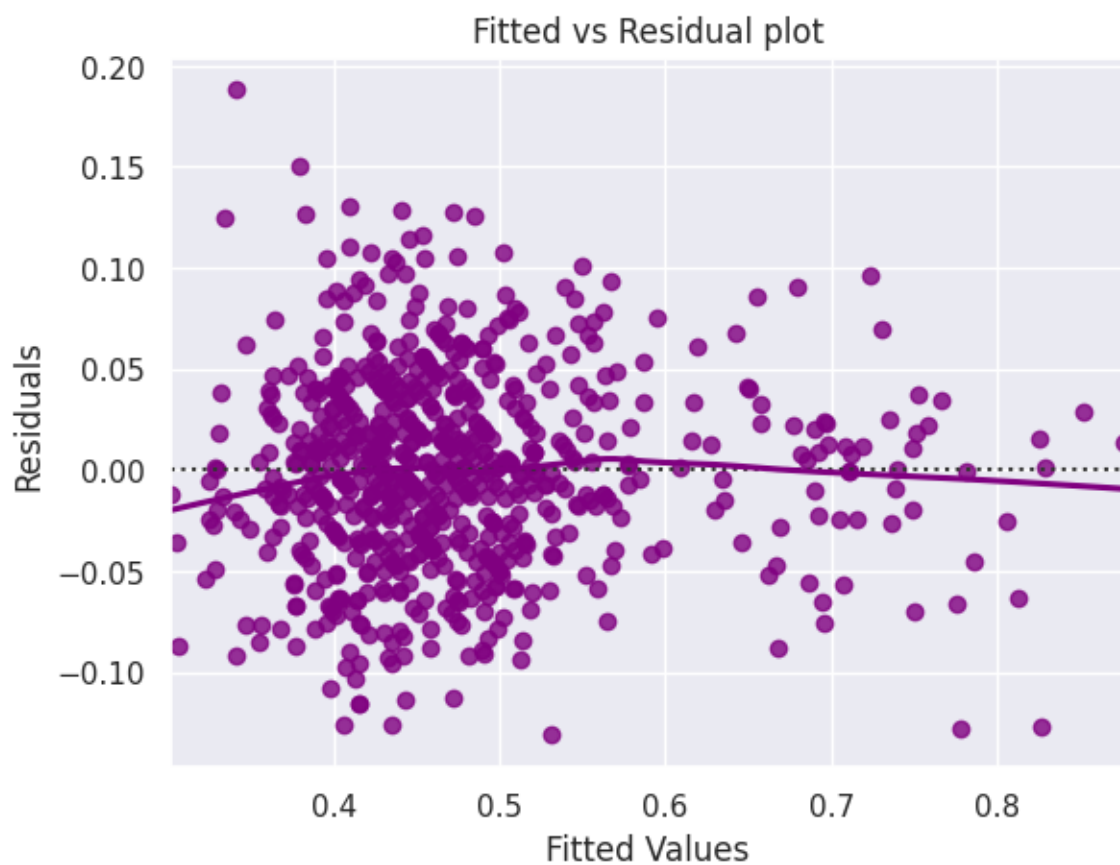


Fig 17. Fitted vs residual plot

Observations:

- The scatter plot shows the distribution of residuals (errors) vs fitted values (predicted values).
- We see no pattern in the plot above. **Hence, the assumptions of linearity and independence are satisfied.**

3. TEST FOR NORMALITY

- Residuals, should be normally distributed. Non-normality suggests that there are a few unusual data points that must be studied closely to make a better model.
- Histogram, Q-Qplot and a Shapiro-wilk's test help understand the normality of the data.

Histogram for Normality:

- The histogram of residuals have a bell shape and slightly right skewed.

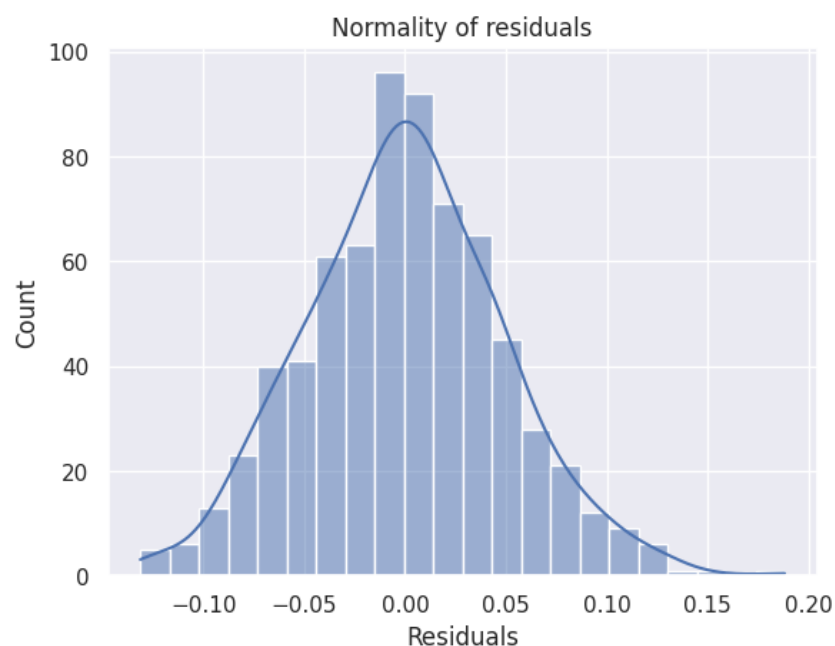


Fig 18. Histogram of Normality of Residuals

Q-Q PLOT FOR NORMALITY:

- Let's check the Q-Q plot.

- The residuals more or less follow a straight line except for the tails.

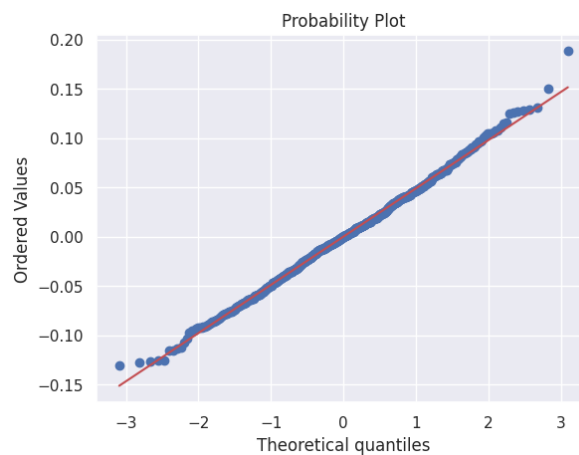


Fig 19. Q-Q plot for normality

SHAPIRO-WILK'S TEST FOR NORMALITY:

Null and alternative hypothesis for Shapiro-Wilks test:

- Null hypothesis: Residuals are normally distributed
- Alternate hypothesis: Residuals are not normally distributed
- Since $p = 0.3109$ is greater than 0.05 , fail to reject H_0 , hence residuals are likely normally distributed
- **So, the Normality assumption is satisfied.**

4. TEST FOR HOMOSCEDASTICITY

- If the variance of the residuals is symmetrically distributed across the regression line, then the data is said to be homoscedastic.
- The residual vs fitted values plot can be looked at to check for homoscedasticity.
- Since $p\text{-value} > 0.05$, we can say that the residuals are homoscedastic. So, this assumption is satisfied.

Predictions on test data:

- Now that we have checked all the assumptions of linear regression and they are satisfied, let us go ahead with prediction.
- We can observe here that our model has returned pretty good prediction results, and the actual and predicted values are comparable

FINAL MODEL OBSERVATIONS:

```
=====
                        OLS Regression Results
=====
Dep. Variable:          views_content    R-squared:                0.789
Model:                  OLS              Adj. R-squared:          0.786
Method:                 Least Squares    F-statistic:             233.8
Date:                   Sun, 15 Jun 2025  Prob (F-statistic):      7.03e-224
Time:                   16:13:29         Log-Likelihood:          1120.2
No. Observations:       700             AIC:                    -2216.
Df Residuals:           688             BIC:                    -2162.
Df Model:               11
Covariance Type:        nonrobust
=====
                        coef    std err          t      P>|t|      [0.025    0.975]
-----
const                   0.0747     0.015     5.110     0.000     0.046     0.103
visitors                0.1291     0.008    16.440     0.000     0.114     0.145
major_sports_event     -0.0606     0.004   -15.611     0.000    -0.068    -0.053
views_trailer           0.0023    5.5e-05   42.414     0.000     0.002     0.002
dayofweek_Monday        0.0321     0.012     2.731     0.006     0.009     0.055
dayofweek_Saturday      0.0570     0.007     8.042     0.000     0.043     0.071
dayofweek_Sunday        0.0344     0.008     4.456     0.000     0.019     0.050
dayofweek_Thursday      0.0154     0.007     2.307     0.021     0.002     0.029
dayofweek_Wednesday     0.0465     0.004    10.532     0.000     0.038     0.055
season_Spring            0.0226     0.005     4.259     0.000     0.012     0.033
season_Summer            0.0434     0.005     8.112     0.000     0.033     0.054
season_Winter            0.0282     0.005     5.362     0.000     0.018     0.039
=====
Omnibus:                 3.254    Durbin-Watson:           1.996
Prob(Omnibus):            0.196    Jarque-Bera (JB):         3.077
Skew:                     0.139    Prob(JB):                 0.215
Kurtosis:                 3.168    Cond. No.                  662.
=====
```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

- The model is able to explain ~78% of the variation in the data
- The train and test RMSE and MAE are low and comparable. So, our model is not suffering from overfitting
- The MAPE on the test set suggests we can predict within 9.17% of the anime ratings
- Hence, we can conclude the model *olsmodel_final* is good for prediction as well as inference purposes.

CONCLUSIONS AND RECOMMENDATIONS:

1. The model can explain ~78% of the variation in the data and within 9.17% of the ott data on the test data, this indicates that the model is good for prediction as well as inference purposes
2. If the number of visitors increases by one unit, then its Ott content views increase by 0.1287 units, all other variables held constant
3. If the sports event increases by one unit, then its content viewership decreases by 0.0606 units, all other variables held constant. Hence this event should be considered while releasing new content.
4. Since the content views on Saturday are high, it is preferable to release more content on this day.
5. As the ott views increase with an increase in the number of people watching it, the company can consider better marketing strategies to reach more audience.
6. Since the content views are more in summer season, hence better and more content can be launched during this season.